PLANT DISEASE DETECTING ROBOT

ROBOTICS AND INDUSTRIAL AUTOMATION

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We, DAIVEEK LOK**ESH (1MJ24CS0**37**),** DHANUSH **REDDY (1MJ24CS0**42**),** GAGAN C **(1MJ24**IS023**)** students of Fourth Semester B.E., Department of Electronics and Communication Engineering, MVJ College of Engineering, Bengaluru, hereby declare that the AEC Level II project titled '**PLANT DISEASE DETECTING ROBOT**' has been carried out by us and submitted in partial fulfilment for the award of Degree of **Bachelor of Computer Science of Engineering** during the year 2025-2026.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

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ABSTRACT

Agriculture plays a major role in the economy and food production of many countries. One of the biggest challenges faced by farmers is the early detection of plant diseases, as diseases can rapidly reduce crop quality and yield if not identified in time.

Traditional disease monitoring methods require continuous manual inspection by experts, which is time-consuming, costly, and less efficient for large agricultural fields. To overcome these limitations, the proposed project “Plant Disease Detecting Robot” is designed as an intelligent robotic system capable of automatically identifying plant diseases and assisting farmers in crop monitoring.

The robot is equipped with sensors, a camera module, and an image-processing system to capture images of plant leaves and analyze them for symptoms of diseases. Using machine learning and image recognition techniques, the system compares captured images with a trained dataset to detect diseases accurately.

The robot can move autonomously through agricultural fields, monitor plant health in real time, and notify farmers about detected diseases through a display or wireless communication module.

The proposed system helps reduce manual effort, increases detection accuracy, and enables early treatment of infected plants, thereby improving crop productivity and minimizing economic losses. In addition, the robot can operate continuously in different environmental conditions, making it a cost-effective and efficient solution for smart farming applications. This project demonstrates the integration of robotics, artificial intelligence, and IoT technologies in modern agriculture to support sustainable and precision farming practices.

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INTRODUCTION

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INTRODUCTION

Agriculture is one of the oldest and most important occupations in the world. It plays a vital role in providing food, employment, and raw materials for industries. The growth of the agricultural sector directly influences the economic development of many countries. With the increasing global population, the demand for food production is also increasing rapidly. To meet this growing demand, farmers must improve crop productivity while reducing losses caused by diseases, pests, and environmental conditions.

Plant diseases are one of the major factors affecting agricultural productivity. Diseases caused by fungi, bacteria, viruses, and pests can spread rapidly and damage crops severely if not identified at an early stage. Infected plants often show symptoms such as leaf spots, discoloration, wilting, and abnormal growth. If proper action is not taken immediately, diseases can reduce both the quality and quantity of crop yield, resulting in huge economic losses for farmers.

Traditionally, plant disease detection is carried out manually by farmers or agricultural experts through visual inspection. Although manual monitoring is widely practiced, it has several disadvantages. It requires continuous observation, consumes a large amount of time, and depends heavily on the experience of the observer. In large agricultural fields, manual inspection becomes difficult and inefficient. Sometimes diseases are identified only after they have spread extensively, making treatment less effective.

The rapid development of modern technologies such as robotics, artificial intelligence, image processing, machine learning, and Internet of Things (IoT) has created new opportunities for smart agriculture. Automated agricultural systems can monitor crops continuously, analyze plant conditions accurately, and assist farmers in making quick decisions. These intelligent systems help improve productivity, reduce labor costs, and support sustainable farming practices.

The Plant Disease Detecting Robot is an innovative smart farming solution developed to automate the process of crop monitoring and disease detection. The robot is designed to move autonomously through agricultural fields and capture images of plant leaves using a camera module. These images are processed using image-processing techniques and machine learning algorithms to identify diseases accurately.

The robot consists of several important components including a microcontroller, camera module, sensors, motor driver, DC motors, wireless communication modules, and power supply system. The microcontroller acts as the brain of the system and controls the movement of the robot as well as communication between different hardware modules. Sensors are used to monitor environmental conditions such as temperature and humidity, which are important factors influencing plant diseases.

Image processing plays a significant role in disease detection. The captured leaf images undergo preprocessing operations such as filtering, segmentation, resizing, and feature extraction. Important features such as leaf texture, color variations, and spot patterns are analyzed to determine whether the plant is healthy or infected. Machine learning algorithms compare these extracted features with trained datasets and classify the disease type.

The “Plant Disease Detecting Robot” is an advanced agricultural robotic system developed to solve this problem using modern technologies such as robotics, image processing, machine learning, and IoT. The robot is capable of moving through farmland, capturing images of plant leaves using a camera, and analyzing them to identify diseases automatically. By comparing the captured images with a trained dataset, the system can recognize disease symptoms and provide accurate results in real time.

The robot reduces the dependency on manual inspection and helps farmers take preventive actions quickly. It improves efficiency, reduces crop loss, and supports precision farming techniques.

1.1 MOTIVATION

- Early detection of plant diseases reduces crop loss.
- Automated monitoring minimizes manual effort.
- Smart farming improves agricultural productivity.
- Machine learning enables accurate disease prediction.
- Robotic systems support modern precision agriculture.

1.2 OBJECTIVES

- To design a robot capable of detecting plant diseases automatically.
- To capture plant images using a camera module.
- To identify diseases using image processing and machine learning.
- To provide real-time alerts and monitoring.
- To improve farming efficiency and crop productivity.

CHAPTER 2 LITERATURE SURVEY

CHAPTER 2 LITERATURE SURVEY

[1] Sharma, P., & Gupta, R. (2021) proposed an automated crop monitoring system using IoT sensors and image processing for plant disease detection. The system improved disease identification accuracy and enabled real-time monitoring in agricultural fields.

[2] Kumar, S., & Singh, R. (2020) developed a machine learning–based plant disease detection model using convolutional neural networks. Their work focused on leaf image classification and achieved high accuracy in identifying crop diseases.

[3] Patel, M., & Desai, K. (2019) designed a robotic agricultural monitoring system integrated with wireless communication. The robot was capable of moving through crop fields and capturing images for disease analysis.

[4] Ahmed, T., & Das, P. (2022) proposed a smart farming solution using IoT and AI technologies for monitoring crop health. Their system improved productivity by enabling early disease diagnosis.

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[6] Kim, J., Park, S., & Lee, D. (2020) proposed a vision-based crop monitoring system using OpenCV and machine learning techniques. The system analyzed leaf texture and color variations to detect diseases in tomato plants.

[7] Rossi, F., & Conti, P. (2022) developed an IoT-enabled robotic monitoring system for precision agriculture. The robot transmitted crop condition data through wireless communication networks for real-time monitoring.

[8] Zhao, L., Wang, Y., & Chen, H. (2019) presented a survey on computer vision techniques for agricultural disease detection. The study compared image segmentation and classification algorithms used in smart farming applications.

[9] Singh, R., & Mehta, P. (2021) designed an autonomous agricultural robot capable of monitoring crops and identifying infected regions using machine learning algorithms

[10] Garcia, M., & Torres, A. (2020) analyzed sensor-based environmental monitoring systems for precision agriculture. Their work focused on temperature and humidity monitoring for disease prediction

[11] Sato, K., & Yamamoto, T. (2023) proposed a deep learning–based leaf disease recognition model for smart agriculture applications. The system achieved high classification accuracy using CNN architectures..

[12] Ali, A. S., Mansuri, A., & Siddique, M. M. (2024) developed a smartphone-integrated agricultural monitoring robot capable of identifying diseases and sending reports to farmers through mobile applications.

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[14] D'Souza, R., & Fernandes, J. (2023) proposed an IoT-based smart farming framework that combined environmental sensors and image-processing systems for disease prediction.

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[16] Reddy, P., & Kumar, V. (2022) developed a CNN-based disease prediction model capable of identifying multiple crop diseases from leaf images with high accuracy.

[17] Kobzarev, Banerjee, S., & Chatterjee, A. (2022) presented a robotic vehicle for crop monitoring using wireless communication and sensor networks. The robot improved field coverage and reduced manual inspection.

CHAPTER 3 SYSTEM

DESIGN

CHAPTER 3 SYSTEM DESIGN

3.1 EXISTING SYSTEM

Several plant disease detection systems have been developed over the years to improve agricultural productivity and reduce crop losses. Traditional methods mainly depend on visual observation by farmers or agricultural experts. In these methods, farmers inspect plant leaves manually to identify disease symptoms such as discoloration, leaf spots, fungal growth, and abnormal patterns.

Although manual monitoring methods are simple, they have several limitations. Large agricultural fields require continuous inspection, which consumes significant time and labor. Disease identification also depends on the experience and knowledge of the observer. In many cases, diseases are detected only after they spread extensively across the field.

To overcome these problems, researchers introduced image-processing techniques for automated disease detection. Stationary camera systems were developed to capture images of plant leaves and analyze disease symptoms using computer vision algorithms. These systems improved detection accuracy but had limitations in terms of mobility and field coverage.

Drone-based monitoring systems were later introduced for aerial crop inspection. Drones capture images of large agricultural fields and identify infected areas using image-processing software. Although drone systems provide better coverage, they are expensive, require skilled operation, and consume significant power.

IoT-based agricultural monitoring systems are also widely used in modern farming. These systems use environmental sensors to monitor parameters such as humidity, soil moisture, and temperature. Sensor data helps farmers understand crop conditions and predict possible disease growth. However, many IoT systems lack image-based disease identification capabilities.

Machine learning and deep learning approaches have significantly improved disease classification accuracy. Convolutional Neural Networks (CNNs) are widely used for leaf image classification because they can identify disease patterns automatically. These systems provide high accuracy but often require high computational resources and large training datasets.

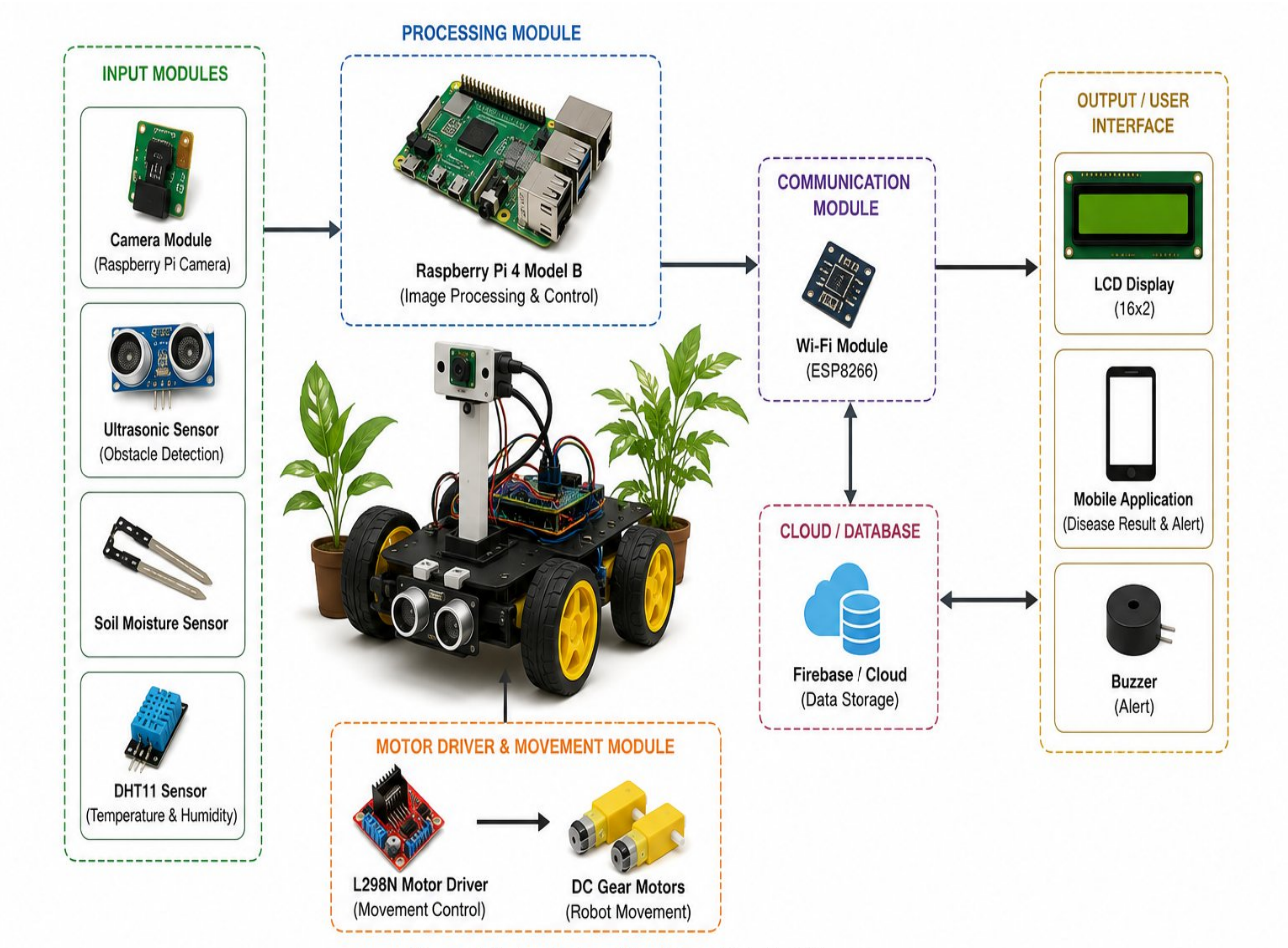

Figure 3.1: System Design of Plant Disease Detecting Robot

Fig 3.1 System Design

3.2 PROPOSED SYSTEM

The proposed Plant Disease Detecting Robot is developed to provide an automated and intelligent solution for monitoring crop health in agricultural fields. The robot integrates advanced technologies such as image processing, machine learning, robotics, IoT, and wireless communication to improve disease detection accuracy and reduce human effort.

The robot consists of a movable robotic platform powered by DC motors and controlled using a motor driver module. A camera module is mounted on the robot to capture high-quality images of plant leaves. These images are analyzed using image-processing algorithms and machine learning models trained on different plant disease datasets.

The microcontroller acts as the central processing unit of the system. It controls robot movement, sensor operation, communication modules, and image acquisition processes. The robot is capable of moving autonomously between crop rows and continuously monitoring plant conditions.

Environmental sensors such as humidity and temperature sensors are included to collect additional data related to plant health. Since environmental conditions influence disease growth, sensor readings help improve the reliability of disease prediction.

The machine learning model used in the system is trained using a large number of diseased and healthy plant leaf images. Feature extraction techniques analyze characteristics such as color variations, texture differences, and leaf spot patterns. Classification algorithms identify the disease category with high accuracy.

The proposed robot offers several advantages compared to existing systems:

- Autonomous field monitoring.
- Early and accurate disease detection.
- Reduction in labor costs.
- Real-time crop monitoring.
- Wireless communication and alerts.
- Support for smart farming applications.
- Improved agricultural productivity.

The system is designed to be low-cost, scalable, and easy to use for farmers. It can be implemented in different types of agricultural environments and extended further with cloud computing and advanced AI-based analysis.

The proposed Plant Disease Detecting Robot is designed as an autonomous agricultural robot capable of monitoring crops and identifying plant diseases in real time. The robot is equipped with a camera module, sensors, microcontroller, motor driver, and wireless communication system.

The camera captures images of plant leaves while the robot moves through the agricultural field. These images are processed using machine learning and image-processing techniques to identify diseases based on color, texture, and leaf patterns. The system compares the captured image with a trained dataset and determines whether the plant is healthy or infected. The robot uses motors and wheels for movement and navigation across the field. A Raspberry pi 4 model B microcontroller controls the overall operation of the robot. Wireless communication modules allow the robot to send alerts and disease reports directly to farmers.

CHAPTER 4

MATERIALS AND METHODOLOGY

CHAPTER 4 MATERIALS AND METHODOLOGY

1.1. MATERIALS FOR DEVICE

1.1.1.Raspberry pi 4 model B

The Raspberry pi 4 model B and ESP32 microcontrollers are used as the main controlling units of the Plant Disease Detecting Robot. These microcontrollers process sensor data, control robot movement, manage communication modules, and coordinate overall system operation.

- Easy programming using Raspberry Pi
- Support for wireless communication.
- Low power consumption.
- Compatibility with various sensors and modules.
- Cost-effective implementation.
- High reliability for embedded applications.

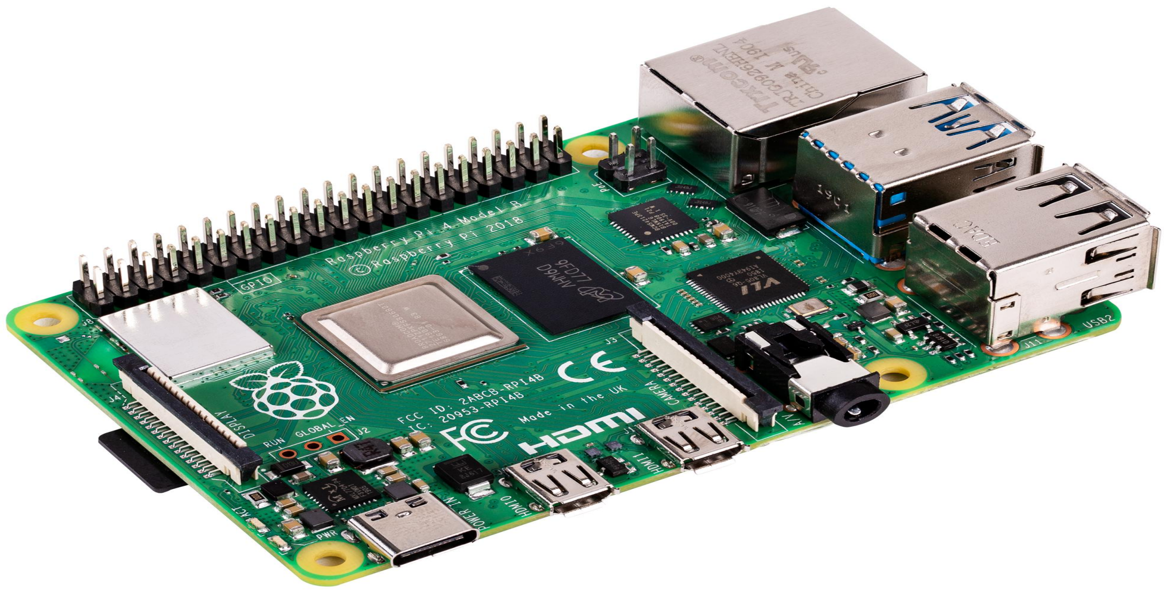

Fig 4.1.1 Raspberry Pi 4 model B

1.1.2.Mobile remote chassis

- The mobile remote chassis is the base structure of the robot.
- It supports all components such as the Raspberry Pi, camera, sensors, battery, and motors.
- It enables the robot to move through agricultural fields for plant monitoring and disease detection.
- A 4-wheel drive (4WD) chassis is preferred for better stability and movement on uneven surfaces.
- The chassis is commonly made of acrylic, aluminum, or metal materials.
- DC geared motors are used to provide controlled movement and sufficient torque.
- Wheels are attached to the motors for forward, backward, and turning movements.
- A motor driver module such as the L298N Motor Driver Module is used to control the motors.

Fig 4.1.2 Mobile remote chasis

1.1.3.Jumper Wires

- **Circuit Prototyping: Jumper wires are widely used on breadboards to quickly build and test circuits without permanent connections.**

- **Connecting Components: They are used to connect components like sensors, LEDs, resistors, and modules to microcontrollers such as Arduino.**

- **Temporary Testing and Debugging: During circuit testing, jumper wires help in making temporary connections that can be easily changed or corrected.**

- **Educational and Laboratory Use: They are commonly used in labs and projects for learning electronics due to their simplicity and ease of use.**

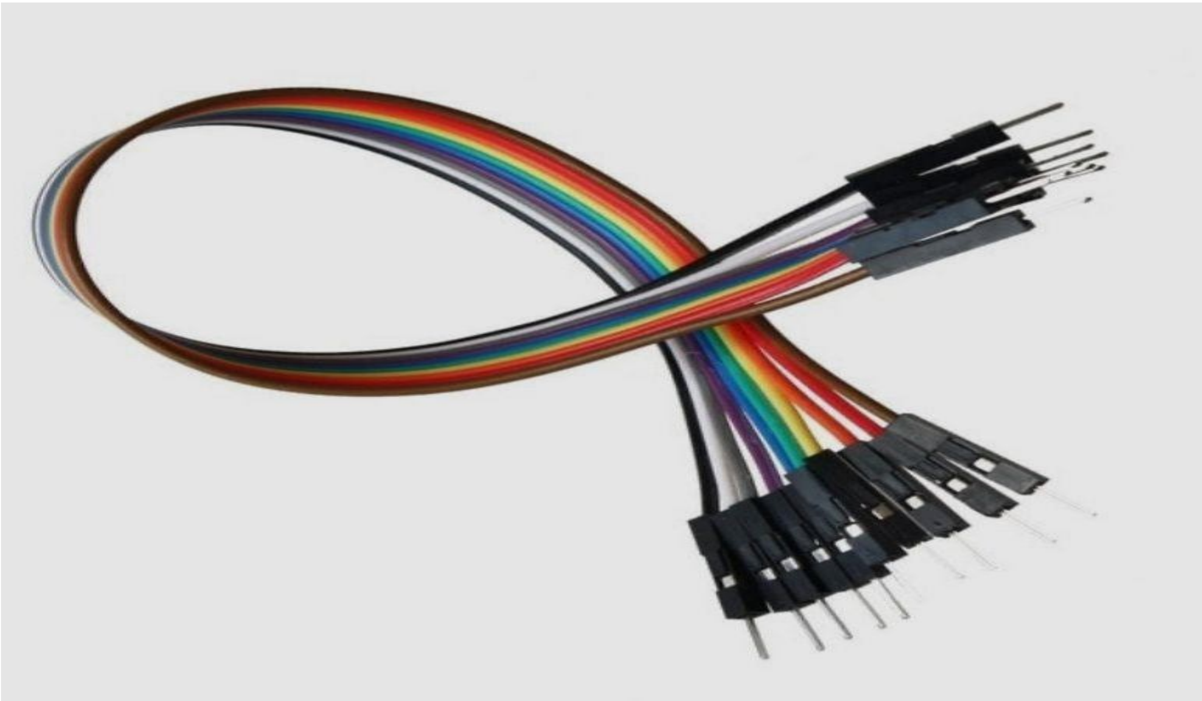

Fig 4.1.3 Jumper Wires

1.2.SOFTWARE REQUIREMENTS

1.2.1.Arduino IDE (for ESP32/Arduino)

- Provides a user-friendly environment to write, compile, and upload code to Arduino and ESP32 boards.
- Includes extensive libraries for IMU sensors, motor drivers, and wireless communication modules.
- Offers a built-in **Serial Monitor** for debugging sensor readings and gesture signals.
- Cross-platform support (Windows, macOS, Linux) makes it accessible for most users.
- Open-source and widely supported, ensuring reproducibility and community resources.

1.2.2.Python + MediaPipe (for vision approach)

- Enables camera-based gesture recognition using Google's MediaPipe framework for hand tracking.
- Provides real-time detection of hand landmarks and finger positions for gesture classification.
- Runs on PCs, laptops, or Raspberry Pi, making it flexible for hybrid control setups.
- Integrates easily with communication protocols (MQTT, sockets) to send recognized gestures to the robot.
- Useful for validating IMU-based gestures or adding redundancy in noisy environments.

1.2.3.TensorFlow Lite Micro (for ML on MCU)

- Allows deployment of lightweight machine learning models directly on microcontrollers.
- Enables advanced gesture classification beyond simple threshold-based rules.
- Optimized for low-power, resource-constrained devices like Arduino Nano or ESP32.
- Supports real-time inference with minimal latency for responsive robot control.
- Facilitates training on custom gesture datasets for personalized recognition.

1.2.4.MQTT / Socket Communication

- Provides lightweight communication between glove and robot using publish/subscribe architecture.
- Supports multi-node setups, enabling integration with IoT devices or cloud services.
- Ensures reliable data transfer with low bandwidth requirements.
- Socket communication allows direct peer-to-peer data exchange for real-time control.
- Easily integrates with ROS or Python scripts for advanced robotics frameworks.

1.2.5. Serial Monitor / ROS (optional)

- **Serial Monitor: Debugs sensor readings, gesture signals, and communication packets directly from Arduino IDE.**
    - Provides real-time feedback during calibration and testing phases.
- **ROS (Robot Operating System): Offers modular nodes for sensing, control, and visualization.**
- Enables simulation and visualization tools like RViz and Gazebo for research applications.
    - Facilitates reproducibility and scalability in academic or industrial robotics projects.

1.3.METHODOLOGY

The methodology for developing the gesture-controlled robot begins with the design of the gesture acquisition system on the wearable side. An IMU sensor such as the MPU6050 or MPU9250 is mounted on a glove or wrist strap to capture accelerometer and gyroscope data. This raw data reflects the tilt, rotation, and orientation of the user's hand. To ensure accuracy, the sensor readings are calibrated and filtered using techniques like complementary or Kalman filtering, which reduce noise and drift. The processed signals are then mapped to specific gesture commands, such as forward tilt for forward movement or left tilt for turning left.

Once the gestures are identified, the microcontroller on the glove side, typically a Raspberry Pi 4 model B or ESP32, encodes these commands and transmits them wirelessly. Communication is established through modules such as the HC-05 Bluetooth or the ESP32's built-in Wi- Fi/Bluetooth, ensuring reliable packet transfer between the wearable and the robot.

Error-checking mechanisms are incorporated to prevent misinterpretation of signals, and the system is designed to stop the robot safely in case of signal loss. On the robot side, another microcontroller such as an Raspberry pi 4 model B or ESP32 receives the transmitted gesture commands. These commands are decoded and passed to the motor driver, such as the L298N or TB6612FNG, which regulates the voltage and current supplied to the DC geared motors. The motors, mounted on a two-wheel or four-wheel chassis with an additional caster wheel for balance, provide the robot's mobility. The battery pack supplies regulated power to both the microcontroller and the motor driver, with protective circuits to prevent overload or short circuits.

System integration involves both hardware and software coordination. Hardware integration ensures that sensors, communication modules, motors, and power systems are securely mounted and interconnected. Software integration involves programming the microcontrollers to handle sensor acquisition, gesture filtering, wireless communication, and motor control logic. Safety features such as automatic stop on signal loss and overload protection are embedded into the control algorithms to enhance reliability.

The final stage of the methodology focuses on testing and validation. Gesture accuracy is evaluated by checking whether the robot consistently responds to intended hand movements. Response time is measured to ensure minimal latency between gesture input and robot action. Stability tests confirm smooth motion and balance, while range tests assess the reliability of wireless communication over varying distances. Error-handling mechanisms are validated by simulating signal loss or sensor noise to ensure the robot responds safely.

Through this methodology, the gesture-controlled robot achieves touch-free, intuitive, and responsive human–machine interaction. The system is designed to be reproducible, scalable, and adaptable for applications in assistive technology, industrial automation, surveillance, and hazardous environment

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